Visvesvaraya Technological University
BELAGAVI, KARNATAKA.

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ಬೆಳಗಾವಿ, ಕರ್ನಾಟಕ

A PROJECT REPORT ON

"AI FOR SILENT SPEECH RECOGNITION"

Submitted to Visvesvaraya Technological University in partial fulfillment of the requirement for the award of Bachelor of Engineering degree in Artificial Intelligence and Machine Learning.

Submitted by

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Dec 2025

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CERTIFICATE

Certified that the project work entitled "**AI FOR SILENT SPEECH RECOGNITION**" carried out by **Adithi H S (4JN22AI001), Pratheeksha K N (4JN23AI402),** bonafide students of **J N N College of Engineering** in partial fulfillment for the award of Bachelor of Engineering in **Department of Artificial Intelligence and Machine Learning** of the Visvesvaraya Technological University, Belagavi during the year **2025-26**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

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Abstract

Silent Speech Recognition (SSR) using Artificial Intelligence is an emerging research area that focuses on interpreting spoken language from visual cues, particularly lip movements, without relying on acoustic signals. This technology holds significant potential for applications such as assistive communication for speech-impaired individuals, communication in noisy or sensitive environments, and advanced human–computer interaction. This project presents the development of an AI-driven lip-reading system that utilizes deep learning techniques to recognize speech from video data. Convolutional Neural Networks (CNNs) are employed to extract spatial features from lip movements, while Recurrent Neural Networks (RNNs) model the temporal dynamics of visual speech patterns across frames. The system is trained on large-scale video datasets of spoken language, enabling it to learn discriminative visual speech representations. Experimental results demonstrate promising performance in recognizing both isolated words and continuous speech, highlighting the effectiveness of deep learning–based lip-reading approaches and their potential to bridge the gap between non-verbal visual speech and digital communication systems.

Acknowledgements

On the very outset of this project report on "**AI-for Silent Speech Recognition**", we would like to extend our sincere & heartfelt obligation towards all the persons who have helped us in this endeavor. Without their active guidance, help, co-operation & encouragement, we would not have made headway in the project.

We would like to thank our beloved guide Ms. Aaliya Waseem, Assistant Professor, Department of Artificial Intelligence and Machine Learning, Dr. Chetan K R, Professor and Head, Department of Artificial Intelligence and Machine Learning and Dr. Y. Vijaya Kumar, Principal, Jawaharlal Nehru New College of Engineering, Shivamogga for all their support and encouragement.

We are thankful and fortunate enough to get constant encouragement, support and guidance from all teaching staff of the Department of Artificial Intelligence and Machine Learning. Also, we would like to extend our sincere regards to all the non-teaching staff of the Department of Artificial Intelligence and Machine Learning for their timely support.

We are grateful to our institution Jawaharlal Nehru New College of Engineering and the Department of Artificial Intelligence and Machine Learning for imparting us the knowledge with which we can do our best.

Thank you,

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Chapter 1

INTRODUCTION

1.1 Overview

Silent Speech Recognition is an AI-based system that converts lip movements in video into text without using audio signals. The system processes video frames by focusing on the lip region and extracting visual features using deep learning techniques. A combination of Convolutional Neural Networks for spatial feature extraction and Bidirectional LSTM networks for temporal modelling is used, along with the CTC loss function to handle sequence prediction without exact alignment. The trained model predicts the corresponding text from silent speech, and a user-friendly interface allows users to upload or record videos and view the generated output. This project demonstrates an effective solution for communication in noisy or silent environments and has applications in assistive technologies, defence, and human–computer interaction.

1.2 Importance

- Enables speech without sound Silent Speech Recognition allows individuals to communicate without producing audible sound. This is especially useful in noisy environments or situations where speaking aloud is not possible or desirable.
- Supports people with speech impairments The technology helps individuals who cannot speak due to medical conditions. By recognizing lip movements, SSR provides an alternative method for effortless and natural communication.
- In military and defense environments, Silent Speech Recognition enables personnel to communicate without producing audible sound. This is particularly useful during covert missions, as it reduces the risk of detection while ensuring secure and effective communication among soldiers.

1.3 Problem Statement

Conventional speech recognition systems rely heavily on acoustic signals, which significantly limits their effectiveness in noisy environments, restricted-audio scenarios, or situations that require silent communication. These constraints pose challenges in critical domains such as defense, healthcare, and industrial settings, where clear audio capture may not always be feasible. Moreover, individuals with speech or hearing impairments face substantial communication barriers, as most existing digital systems are designed primarily for audio-based interaction. To address these limitations, this project proposes an AI-driven Silent Speech Recognition system that interprets speech using visual cues—specifically lip movements—thereby enabling an inclusive, accessible, and sound-independent communication solution

1.4 Objectives

The primary objective of this project is to design and develop an AI-driven Silent Speech Recognition (SSR) system that accurately interprets spoken language using visual cues, specifically lip movements, without relying on acoustic signals. The detailed objectives of the project are as follows:

1. To study and analyze existing speech recognition and lip-reading techniques to identify their limitations in noisy, silent, and accessibility-critical environments.
2. To collect and preprocess video-based speech data by extracting lip regions and frame-level visual features suitable for deep learning analysis.
3. To design and implement deep learning models using Convolutional Neural Networks (CNNs) for spatial feature extraction from lip movements.
4. To employ Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks to capture temporal dependencies in visual speech sequences.
5. To train and evaluate the proposed model on standard lip-reading datasets for both isolated word and continuous speech recognition.
6. To assess the system's performance using appropriate evaluation metrics and analyze its effectiveness under sound-independent conditions.
7. To demonstrate the applicability of the proposed SSR system as an inclusive communication solution for speech-impaired users and silent communication scenarios.

These objectives collectively aim to establish a robust, accessible, and intelligent silent speech recognition framework that bridges the gap between visual speech and effective digital communication.

1.5 Organization of the report

This report is organized into several chapters, each addressing a specific aspect of the Silent

Speech Recognition project. **Chapter 1** introduces the project, providing the background, motivation, problem statement, and objectives of the proposed system. **Chapter 2** presents a comprehensive literature survey covering existing speech recognition, lip-reading techniques, and related deep learning approaches. **Chapter 3** discusses the requirement analysis, including functional and non-functional requirements, system constraints, and relevant outcome mappings. **Chapter 4** details the design and implementation of the proposed Silent Speech Recognition system, describing data preprocessing, model architecture, and training methodology. **Chapter 5** presents the experimental results along with performance analysis and discussion. Finally, **Chapter 6** concludes the report by summarizing the project outcomes and highlighting potential directions for future enhancements, followed by the references.

Chapter 2

LITERATURE SURVEY

2.1 Lips Reading Using Deep Learning

Lip Reading Using Deep Learning focuses on recognizing spoken words by analyzing lip movements from video data without using audio signals. Deep learning models such as CNNs and LSTMs are used to extract visual features and learn temporal patterns for accurate speech recognition.

Methodology

The proposed methodology employs a deep learning–based lip-reading framework named LipReader++, which integrates 3D Convolutional Neural Networks (3D CNNs) and Transformer architectures to recognize speech from visual inputs alone. Initially, video data from benchmark datasets such as GRID and LRS2 is preprocessed through face detection, lip region extraction, normalization, and resizing to ensure consistent input quality. Data augmentation techniques, including cropping, flipping, and brightness adjustment, are applied to improve robustness and generalization. The extracted lip-region video sequences are passed through a 3D CNN to capture spatial and temporal features of lip movements, which are then modeled using Transformer networks to learn long-range temporal dependencies. The model is trained using a combination of CTC loss and cross-entropy loss, optimized with the Adam optimizer and regularized using dropout and L2 regularization to prevent overfitting. Performance is evaluated using metrics such as accuracy and word error rate (WER), demonstrating the effectiveness of the approach in silent speech recognition under diverse conditions.

Contributions

The paper introduces LipReader++, a deep learning–based lip-reading model that combines 3D CNNs and dual-Transformer architectures to effectively capture spatio-temporal features from silent video. With robust preprocessing and data augmentation, the model achieves superior performance on benchmark datasets and demonstrates strong potential for real-world silent speech recognition applications.

Advantages

- Effective in noisy or audio-restricted environments: Operates reliably when audio input is unclear, unavailable, or prohibited, ensuring consistent speech recognition.
- Improves accessibility for individuals with speech or hearing impairments: Converts lip movements directly into text, enabling inclusive and assistive communication.
- High recognition accuracy: Utilizes advanced CNN and Transformer architectures to capture subtle and complex lip movements, improving word- and sentence-level prediction accuracy.
- Supports silent and real-time communication: Enables secure, hands-free, and real-time interaction, making it suitable for defense, healthcare, assistive technologies, and human– computer interaction.

Limitations

- Sensitivity to visual quality: The model requires high-quality, frontal-view video input and its performance degrades under poor lighting, low resolution, side-angle views, or partial occlusion of the lips (e.g., masks or objects).
- Difficulty with visually similar phonemes: Lip movements corresponding to similar sounds such as b, p, and m are hard to distinguish visually, leading to recognition errors.
- Reduced performance with fast speech: Rapid lip movements during high-speed speech are challenging to capture accurately, resulting in decreased recognition accuracy.
- High computational requirements: The use of 3D CNNs and Transformer architectures demands significant computational resources, which can limit real-time deployment on low-end or resource-constrained devices

2.2 A Comprehensive Review of Recent Advances in Deep Neural Networks for Lipreading with Sign Language Recognition

It provides a comprehensive review of recent advances in deep neural network techniques applied to lipreading and sign language recognition.

It highlights key architectures and methods that improve visual speech and gesture understanding for enhanced human– computer interaction.

Methodology

This study follows a structured and systematic review methodology to examine recent advancements in deep neural network–based lipreading with sign language recognition. Relevant research articles published mainly from 2010 onwards were collected from reputed digital libraries, with a strict focus on deep learning–driven approaches. The reviewed studies are analyzed and organized according to the standard lipreading framework, which includes video acquisition, preprocessing (face detection, lip localization, region-of-interest extraction, and normalization), feature extraction, temporal modeling, and classification. The methodology critically compares traditional handcrafted features with deep visual features learned using architectures such as Convolutional Neural Networks (CNNs), 3D-CNNs, Recurrent Neural Networks (LSTM and GRU), Temporal Convolutional Networks, and Transformer-based models. In addition, widely used benchmark datasets, covering multiple languages and recognition levels (digits, words, and sentences), are reviewed along with evaluation metrics such as accuracy and word error rate. The integration of lipreading with sign language recognition is also analyzed to highlight existing challenges, limitations, and emerging research trends, providing a comprehensive foundation for future work in multimodal and assistive communication system

Contributions

This paper provides a comprehensive review of recent deep learning–based approaches for lipreading and sign language recognition, highlighting their integration in multimodal communication systems. It analyzes modern neural architectures, commonly used datasets, feature extraction methods, and evaluation metrics, while comparing traditional and deep learning techniques. The study also identifies key challenges such as data scarcity, variability, and real-time constraints, and outlines future research directions to guide further advancements in assistive and inclusive communication technologies.

Advantages

- Provides a comprehensive and up-to-date review of deep learning techniques for lipreading combined with sign language recognition, covering recent advancements in the field.

- Offers a clear comparison of neural architectures and feature extraction methods, helping
- researchers understand the strengths of CNNs, RNNs, TCNs, and Transformer-based models.
- Presents an extensive analysis of benchmark datasets and evaluation metrics, which supports
- informed dataset selection for future research.
- Identifies key challenges and future research directions, guiding the development of more
- robust, scalable, and inclusive multimodal communication systems

Limitations

- •The study is limited to deep learning-based approaches and does not include non–deep learning or hybrid traditional methods, which may still be relevant in resource-constrained scenarios.
- Most of the reviewed works rely on controlled or benchmark datasets, limiting insights into real-world performance under varying lighting conditions, occlusions, and spontaneous speech.
- The paper highlights existing datasets but does not provide quantitative experimental validation or unified benchmarking, as it is a survey-based study rather than an implementation-focused analysis.
- Language coverage remains restricted to a few widely studied languages, with limited discussion on low-resource languages and cross-linguistic generalization challenges

2.3 Text Extraction and Translation Through Lip Reading Using Deep Learning

This paper presents a deep learning based lip reading system that extracts spoken text by analyzing lip movements from video input. The extracted text is further translated into different languages, improving communication for speech- and hearing-impaired users.

Methodology

The proposed system follows a deep learning–based pipeline for text extraction and translation through lip reading. Initially, video sequences are captured and preprocessed to detect and isolate the lip region using a CNN-based object detection approach.

Spatial features representing lip shape and motion are then extracted from consecutive frames using Convolutional Neural Networks, enabling effective modeling of visual speech patterns. These features are fed into a recurrent sequence modeling architecture, such as RNNs/LSTMs, to capture temporal dependencies and generate corresponding textual representations of the spoken content. Finally, the extracted text is passed through a machine translation module to translate the recognized speech into the target language, enabling accurate transcription and multilingual communication without relying on audio signals

Contributions

The paper presents a deep learning–based system for extracting and translating text through lip reading without relying on audio signals. The methodology employs Convolutional Neural Networks to detect the lip region and extract spatial features from video frames. Recurrent Neural Networks are then used to model temporal dependencies in lip movements and convert them into textual representations. A machine translation module further translates the recognized text into the target language. The proposed system is effective in noisy or silent environments and supports multilingual communication. This approach improves accessibility for hearing-impaired individuals and expands the applicability of visual speech recognition systems.

Advantages

- Enables accurate speech recognition without relying on audio input, making it effective in noisy, silent, or audio-restricted environments.
- Improves accessibility and communication support for individuals with hearing impairments through visual-based speech understanding.
- Integrates machine translation with lip reading, allowing multilingual text output and reducing language barriers.
- Utilizes deep learning architectures (CNNs and RNNs) to capture both spatial and temporal lip movement features, resulting in improved robustness and recognition accuracy compared to traditional methods

Limitations

- The system's performance degrades under poor lighting conditions, extreme head
- poses, or occlusions such as masks or partial face coverage.

- Accuracy may be affected by variations in accents, speaking styles, and fast lip movements, leading to ambiguity in visually similar phonemes.
- The model requires large, diverse, and well-annotated video datasets for effective training, which limits scalability to low-resource languages.
- Deep learning–based architectures are computationally intensive, making real-time deployment challenging on resource-constrained devices

2.4 An Automatic Lip Reading for Short Sentences Using Deep Learning Nets

This paper proposes an automatic lip reading system for recognizing short sentences using deep learning networks. The model learns visual speech patterns from lip movements to accurately predict spoken sentences without relying on audio input.

Methodology

The proposed methodology presents an automatic visual speech recognition system designed to classify short spoken sentences using deep learning. Initially, input videos are captured and segmented into individual sentence-level clips, from which frames are extracted for further processing. Face detection is performed on each frame using the Viola–Jones algorithm to isolate the facial region from complex backgrounds. Subsequently, facial landmark detection is applied to identify 68 key facial points, where landmarks 48 to 68 correspond to the lip region. A binary mask is constructed based on these landmarks to accurately extract the lip area, effectively handling challenges such as facial hair occlusions. To improve visual quality, the extracted lip images undergo contrast enhancement by converting the color space from RGB to Lab and adjusting the luminance component. Finally, the enhanced lip frames are resized and classified using two pre-trained convolutional neural network models, namely AlexNet and VGG-16, enabling end-to-end sentence-level lip-reading based solely on visual information.

Contributions

This paper makes a significant contribution by proposing a complete and effective automatic lip-reading framework for recognizing short English sentences using only visual information. It introduces a robust lip region extraction method that combines Viola–Jones face detection with 68-point facial landmark localization and binary mask generation, enabling accurate lip isolation even in challenging cases such as facial hair occlusion.

In addition, the study incorporates a dedicated lip enhancement stage using contrast adjustment in the Lab color space to improve visual feature quality prior to classification. The paper further provides a comparative evaluation of two well-known deep learning architectures, Alex Net and VGG- 16, demonstrating that Alex Net achieves superior performance for short sentence classification. Through extensive experiments on a multi-speaker dataset, the work validates the effectiveness of the proposed pipeline and establishes a practical benchmark for sentence- level visual speech recognition systems.

Advantages

- Enables accurate visual-only speech recognition, making the system suitable for noisy environments and for individuals with hearing or speech impairments.
- Uses a robust lip region extraction strategy based on Viola-Jones face detection and 68-point facial landmarks, which effectively removes background clutter and non-relevant facial areas.
- The binary mask-based lip extraction handles real-world challenges such as mustache and beard occlusion, improving reliability across male and female speakers.
- Incorporation of a lip enhancement stage using contrast adjustment improves image quality and feature visibility, leading to better classification performance.
- Experimental comparison shows that Alex Net achieves high classification accuracy (90%) for short sentences, demonstrating the effectiveness of deep CNNs for sentence-level lip-reading tasks.

Limitations

- The system is sensitive to variations in lighting conditions and head poses, which can negatively affect lip detection and classification accuracy.
- Performance may degrade for individuals with unique facial characteristics or uncommon speech patterns, limiting robustness across diverse users.
- The approach relies on high-quality, frontal-view video input, making it less effective under low-resolution or unconstrained recording conditions.
- The deep learning models require a large, annotated training dataset, and may not generalize well to unseen data or different recording environments.
- The system is evaluated only on short, predefined English sentences, restricting its applicability to continuous speech or multilingual scenarios

2.5 A Deep Dive into Existing Lip Reading Technologies

This paper provides an in-depth analysis of current lip reading technologies, exploring various approaches and their performance. It highlights the strengths and limitations of existing methods to inform future research in visual speech recognition.

Methodology

The paper presents a comprehensive deep learning–based methodology for visual speech recognition (lip reading) that converts silent video input into textual output through a structured multi-stage pipeline. Initially, video data from standard lip-reading datasets such as the GRID corpus is collected and preprocessed by removing noise, normalizing frames, and extracting the lip region of interest (ROI). The preprocessed videos are then decomposed into sequential frames, from which spatio-temporal features representing lip shape, motion, and dynamics are extracted using 3D Convolutional Neural Networks (3D-CNNs). These features are passed through TimeDistributed layers to preserve temporal consistency and subsequently modeled using Bidirectional Long Short-Term Memory (Bi-LSTM) networks to capture both past and future contextual dependencies in lip movements. To align variable-length visual sequences with output text without requiring frame-level annotations, the model is trained using the Connectionist Temporal Classification (CTC) loss function, optimized with the Adam optimizer and supported by learning rate scheduling and checkpointing for stable training. Optionally, language models are integrated to resolve ambiguities and improve contextual accuracy. Finally, the system outputs predictions at phoneme, word, or sentence level, and performance is evaluated using metrics such as Word Error Rate (WER), Character Error Rate (CER), and BLEU score, demonstrating the effectiveness of the proposed visual speech recognition pipeline.

Contribution

This paper makes a significant contribution by presenting a structured and unified deep learning–based framework for visual speech recognition that consolidates recent advancements in lip-reading technologies into a coherent end-to-end pipeline. It systematically integrates spatio-temporal feature extraction using 3D Convolutional Neural Networks with Bidirectional LSTM networks to effectively capture complex lip movements and long-term temporal dependencies, while employing Connectionist Temporal Classification (CTC) loss to handle sequence alignment without requiring precise frame-level annotations.

The study also provides a detailed methodological breakdown, practical implementation using the GRID dataset, and a comprehensive evaluation using standard metrics such as WER, CER, and BLEU. In addition to proposing an efficient architecture, the paper offers an extensive survey of existing lip-reading approaches, highlighting their strengths and limitations, thereby serving as a valuable reference for researchers and practitioners aiming to develop robust, scalable, and application-ready silent speech recognition systems.

Advantages

- Utilizes an end-to-end deep learning architecture combining 3D Convolutional Neural Networks and Bidirectional LSTM layers, enabling effective extraction of spatio-temporal lip movement features and accurate modeling of long-term temporal dependencies.
- Employs Connectionist Temporal Classification (CTC) loss, which eliminates the need for frame-level alignment between video frames and text, thereby simplifying training and improving robustness for variable-length input sequences.
- Demonstrates strong performance using standard evaluation metrics such as Word Error Rate (WER), Character Error Rate (CER), and BLEU score, indicating reliable and consistent visual speech recognition accuracy.
- Provides a comprehensive and modular methodology that can be easily adapted or extended for different datasets, languages, and real-world applications such as assistive communication and noisy-environment interaction.
- Includes a detailed survey and comparative analysis of existing lip-reading techniques, offering valuable insights into current research trends, strengths, and limitations, which supports future advancements in visual speech recognition.

Limitations

- The system's performance is sensitive to video quality and recording conditions, with accuracy degrading under poor lighting, low resolution, non-frontal face views, or partial occlusions such as masks or hand movements.
- The model relies heavily on large, well-annotated datasets such as the GRID corpus, which limits its generalizability to real-world scenarios and to languages or vocabularies not represented in the training data.

- Visually similar lip movements corresponding to different phonemes (e.g., /b/, /p/, and
- /m/) remain difficult to distinguish, leading to ambiguity and potential errors in word- level and sentence-level predictions.
- The deep learning architecture, particularly the use of 3D-CNNs and Bidirectional LSTMs, is computationally intensive and requires high-end hardware, making real-time deployment on resource-constrained devices challenging.
- The framework is primarily evaluated in controlled environments and may struggle to generalize to unconstrained, real-world settings with diverse speakers, accents, speaking speeds, and background variations.

2.6 Lip Net: Sentence-Level Lipreading

LipNet: Sentence-Level Lipreading introduces an end-to-end deep learning model that performs sentence-level lip reading directly from video input without using audio. It uses spatio-temporal CNNs with LSTM and CTC loss to achieve accurate visual speech recognition.

Methodology

LipNet presents an end-to-end deep learning approach for sentence-level lipreading that converts sequences of mouth-region video frames directly into text. The method extracts spatiotemporal visual features using 3D convolutional layers, models long-range temporal dependencies with bidirectional LSTMs, and performs character-level sequence prediction using the Connectionist Temporal Classification (CTC) loss. By avoiding handcrafted features and explicit alignment between video frames and text, the model handles variable-length inputs and outputs in a unified framework. This architecture enables accurate, speaker-independent lipreading at the sentence level and significantly outperforms previous work-level and human baselines.

Contributions

This paper introduces LipNet, the first end-to-end deep learning model capable of performing sentence-level lipreading directly from raw video input, addressing a major limitation of prior approaches restricted to isolated word or phoneme classification. By integrating spatiotemporal convolutional neural networks for visual feature extraction, bidirectional LSTMs for temporal context modeling, and Connectionist Temporal Classification (CTC) for alignment-free sequence prediction.

LipNet learns both visual representations and sequence mappings jointly within a single framework. The model operates at the character level, supports variable-length input and output sequences, and is trained without explicit frame-to-text alignment, making it fully speaker-independent. Extensive evaluation on the GRID corpus demonstrates that LipNet significantly outperforms existing state-of-the-art methods and even experienced human lipreaders, thereby establishing a new benchmark for automated visual speech recognition and advancing the field toward practical, real-world lipreading applications.

Advantages

- Enables sentence-level lipreading using a single end-to-end model, overcoming the limitation of prior methods that were restricted to isolated word or phoneme classification.
- Eliminates the need for handcrafted visual features and manual alignment, as spatiotemporal features and sequence mappings are learned jointly using CTC loss.
- Effectively captures both spatial mouth movements and temporal dynamics through spatiotemporal CNNs combined with bidirectional LSTMs, leading to higher recognition accuracy.
- Supports variable-length input videos and output text sequences, making the approach flexible and robust for real-world scenarios.

- Demonstrates significantly superior performance compared to existing state-of-the-art methods and experienced human lipreaders on the GRID dataset, establishing a new benchmark in visual speech recognition

Limitations

- The model is evaluated primarily on the GRID corpus, which follows a fixed grammar and limited vocabulary, restricting conclusions about performance on unconstrained, real-world speech.
- LipNet requires high-quality, frontal-view video inputs with clear visibility of the mouth region, and its accuracy may degrade under poor lighting, occlusions, or extreme head poses.
- The approach is computationally intensive, as spatiotemporal convolutions and
- bidirectional LSTMs demand significant training time and hardware resources.

- The system is language- and dataset-dependent, requiring retraining on large, annotated video datasets to generalize to new languages or speaking styles.
- Visual ambiguity between similar-looking phonemes (visemes) can still lead to misclassification errors, particularly when contextual information is insufficient

2.7 Lip2Speech: Lightweight Multi-Speaker Speech Reconstruction with Gabor Features

Lip2Speech presents a lightweight deep learning framework that reconstructs speech from lip movements for multiple speakers. It uses Gabor features to effectively capture visual information, enabling efficient and accurate speech reconstruction from silent video input.

Methodology

The proposed methodology introduces a lightweight visual-to-speech reconstruction framework that generates speech from silent lip movements using Gabor-based visual features. Initially, video frames are extracted from silent speech videos and facial landmarks are detected to localize the mouth region of interest (ROI). Horizontal Gabor filters are then applied to the ROI to suppress irrelevant facial information and emphasize discriminative lip motion patterns. From the Gabor-transformed images, two types of visual representations are derived: (i) low-dimensional Gabor lip images and (ii) compact Gabor lip features capturing geometric and intensity-based mouth dynamics. These visual inputs are temporally modeled using Long Short-Term Memory (LSTM) networks to learn the sequential relationship between lip movements and speech dynamics. The model predicts an encoded auditory spectrogram, which is subsequently decoded to reconstruct the speech waveform using an auditory spectrogram inversion process. Two architectures are designed GaborCNN2Speech, which combines minimal CNN layers with LSTM for spatiotemporal modeling, and GaborFea2Speech, which eliminates CNNs entirely by directly using Gabor features—both aiming to reduce model complexity while maintaining robustness. The system is trained and evaluated on the GRID corpus under single-speaker and multi-speaker settings, demonstrating efficient, speaker-independent speech reconstruction without requiring additional speaker identity or textual information.

Contributions

This paper makes a significant contribution to the field of visual speech reconstruction by proposing a novel, lightweight, and speaker-independent framework for reconstructing speech from silent lip movements using Gabor-based visual features. It introduces two efficient models—GaborCNN2Speech and GaborFea2Speech—that replace complex end- to-end deep CNN architectures with interpretable and low-dimensional Gabor lip representations, thereby reducing model complexity, training time, and computational cost. The proposed approach demonstrates that robust multi-speaker speech reconstruction can be achieved without relying on additional speaker identity, audio cues, or textual information, addressing a key limitation of prior works. Extensive experiments on the GRID corpus validate that the proposed models outperform traditional CNN-based baselines in terms of spectrogram accuracy, speech intelligibility, and stability, while offering faster and more transparent speech reconstruction, marking an important advancement toward practical and scalable silent speech systems.

Advantages

- Enables robust multi-speaker speech reconstruction without requiring additional information such as speaker identity, audio input, or textual cues, overcoming a major limitation of existing methods.
- Employs lightweight and computationally efficient architectures by using low-dimensional Gabor features, significantly reducing model complexity, training time, and resource requirements.
- Improves speech intelligibility and reconstruction quality, achieving higher spectrogram correlation, PESQ, and STOI scores compared to conventional end-to-end CNN-based models.
- Offers greater interpretability and transparency, as Gabor-based visual features explicitly capture meaningful lip geometry and motion rather than relying on opaque deep feature representations.
- Demonstrates stable and consistent performance across single-speaker and multi-speaker scenarios, making the approach suitable for practical real-world silent speech applications

Limitations:

- The system relies on accurate lip region detection and clear frontal facial videos, and its performance may degrade under poor lighting conditions, occlusions, or large head pose variations.
- Speech reconstruction quality is limited by the absence of internal articulatory information which cannot be captured from visual lip data alone.
- The models are trained and evaluated on the GRID corpus, a constrained dataset, which may limit generalization to unconstrained, real-world conversational scenarios.
- Although lightweight, the approach still requires temporal sequence alignment and preprocessing, which can affect real-time deployment efficiency.

2.8 Summary

This chapter briefly reviewed existing literature on silent speech reconstruction, covering traditional feature-based methods and recent deep learning approaches. While modern models improve accuracy, most are complex and speaker-dependent. Multi-speaker reconstruction often relies on additional information such as speaker identity or text. These gaps highlight the need for a lightweight, efficient, and speaker-independent approach.

Chapter 3

REQUIREMENT ANALYSIS

This chapter presents a detailed analysis of the requirements for the proposed Silent Speech Recognition (SSR) system. The requirements are specified following the IEEE Software Requirements Specification (SRS) standard to ensure clarity, consistency, and completeness. The chapter also discusses system constraints, mapping with Programme Outcomes (POs) and Programme Specific Outcomes (PSOs), and the project's alignment with Sustainable Development Goals (SDGs).

3.1 Functional requirements

The functional requirements define the core capabilities and services provided by the proposed system.

- **FR-1: Video Input Handling:** The system shall allow users to upload a video containing clearly visible lip movements as input.
- **FR-2: Frame Extraction:** The system shall extract individual frames from the uploaded video at a predefined frame rate.
- **FR-3: Preprocessing of Video Frames:** The system shall preprocess extracted frames by converting them to grayscale and isolating the lip region of interest (ROI) using facial landmark detection.
- **FR-4: Frame Normalization and Resizing:** The system shall normalize and resize video frames to match the input dimensions required by the trained deep learning model.
- **FR-5: Visual Feature Extraction:**The system shall analyze lip movement patterns using a deep learning–based lip-reading model.
- **FR-6: Sequence Learning and Decoding:** The system shall convert extracted visual features into meaningful text using sequence learning techniques and Connectionist Temporal Classification (CTC) decoding.
- **FR-7: Output Display:** The system shall display the predicted text output to the user through a user-friendly web interface.
- **FR-8: Error Handling:** The system shall detect invalid or unsupported video inputs and notify the user with appropriate error messages.

3.2 Non-Functional requirements

Non-functional requirements describe the quality attributes and operational constraints of the system.

- **NFR-1: Performance** The system shall generate prediction results within an acceptable response time to ensure smooth user interaction.
- **NFR-2: Accuracy** The system shall provide reliable prediction accuracy under controlled lighting conditions and frontal video input.
- **NFR-3: Usability:** The system shall provide a simple, intuitive, and accessible user interface suitable for users with minimal technical knowledge.
- **NFR-4: Scalability:** The system shall support multiple user requests with minimal degradation in performance.
- **NFR-5: Reliability:** The system shall operate consistently without unexpected failures during normal usage.
- **NFR-6: Security:**The system shall securely process uploaded videos and generated outputs without unauthorized access or data leakage.
- **NFR-7: Maintainability:** The system shall support easy updates to the model, dataset, or interface without major architectural changes.
- **NFR-8: Compatibility:** The system shall support commonly used video formats and operate on standard computing environments.

3.3 System Constraints

The proposed Silent Speech Recognition system is subject to certain constraints that influence its design and performance. The accuracy of lip-reading depends heavily on video quality, lighting conditions, camera angle, and visibility of the speaker's lips. Occlusions, facial hair, or extreme head movements may reduce recognition accuracy. The system is also constrained by computational resources, as deep learning inference requires adequate processing power. Additionally, the proposed system is intended as an assistive and supportive communication tool and not as a replacement for professional speech therapy or medical diagnosis.

3.4. PO–PSO Mapping

The Silent Speech Recognition project demonstrates strong alignment with both Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) by integrating core engineering principles with advanced Artificial Intelligence techniques to solve a real-world communication problem. The project applies fundamental knowledge of computer science, machine learning, and deep learning (PO1) to address the limitations of conventional audio-based speech recognition systems. Through systematic analysis of visual speech patterns and model performance, the project reflects effective problem analysis and investigation of complex engineering challenges (PO2 and PO4). The design and development of an AI-driven lip-reading system highlight the ability to create innovative and practical solutions using modern tools and frameworks such as CNNs, RNNs, and CTC decoding (PO3 and PO5).

From a societal and ethical perspective, the project supports inclusive communication by assisting individuals with speech or hearing impairments, thereby addressing broader societal needs and reducing inequalities (PO6 and PO8). The exposure to emerging technologies in visual speech recognition and deep learning also promotes continuous learning and adaptability to evolving technological trends (PO11). In terms of Programme Specific Outcomes, the project strongly fulfills **PSO-1** by demonstrating the application of Artificial Intelligence and Machine Learning concepts to interpret human cognition and visual speech patterns. It also satisfies **PSO-2** by showcasing the development of a complete AI-based system using innovative tools and techniques for real-world problem-solving. Overall, the project effectively bridges theoretical knowledge and practical implementation, aligning well with both POs and PSOs.

PO	**Description**	**Relevance to Project**
PO1	Engineering knowledge	Application of AI, ML, and deep learning concepts
PO2	Problem analysis	Identification of limitations in audio-based speech recognition
PO3	Design/development of solutions	Design of an AI-based silent speech recognition system
PO4	Investigation of complex problems	Analysis of visual speech patterns and model performance
PO5	Modern tool usage	Use of CNNs, RNNs, CTC, and deep learning frameworks
PO6	Engineer and society	Assistive communication for speech-impaired individuals
PO8	Ethics	Inclusive and non-invasive communication approach

3.5 Impact of the Project on Sustainable Development Goals (SDGs)

The Silent Speech Recognition project contributes to multiple United Nations Sustainable Development Goals:

- **SDG 3 – Good Health and Well-Being:** Supports communication assistance for individuals with speech impairments.
- **SDG 4 – Quality Education:** Enables inclusive learning environments for hearing- and speech-impaired individuals.
- **SDG 9 – Industry, Innovation, and Infrastructure:** Promotes innovation in AI-driven human–computer interaction systems.
- **SDG 10 – Reduced Inequalities:** Enhances accessibility and reduces communication barriers for marginalized communities.
- **SDG 11 – Sustainable Communities:** Encourages inclusive digital communication technologies for societal well-being.

3.6 Summary

This chapter detailed the functional and non-functional requirements of the proposed Silent Speech Recognition system in accordance with IEEE SRS standards. It discussed system constraints, mapped the project outcomes to relevant Programme Outcomes and Programme Specific Outcomes, and highlighted the project's broader societal impact through alignment with Sustainable Development Goals. These requirements provide a structured foundation for the system design and implementation described in the subsequent chapters.

Chapter 4

DESIGN AND IMPLEMENTATION

4.1 Design Issues

- Variation in Video Quality: Handling differences in resolution, frame rate, lighting, and background conditions that can affect lip visibility and recognition accuracy.
- Lip Region of Interest (ROI) Extraction: Designing a robust method to accurately detect and crop the lip region from each video frame without loss of important visual features.
- Temporal Modeling of Lip Movements: Capturing the sequential nature of lip movements over time and maintaining correct temporal relationships between frames.
- Speaker Variability: Managing variations in lip shape, speaking style, speed, and facial structure across different users.
- Visually Similar Phonemes: Addressing ambiguity caused by similar-looking lip movements for different sounds such as /b/, /p/, and /m/.
- Model Architecture Selection: Choosing an appropriate deep learning architecture that balances accuracy, training time, and computational efficiency.
- Real-Time Inference Constraints: Ensuring the system can provide predictions with minimal latency for practical usability.
- Hardware and Resource Limitations: Designing the system to perform efficiently on machines with limited CPU, GPU, or memory resources.
- Frontend–Backend Integration: Ensuring smooth interaction between the user interface and backend model for seamless data flow and output display.
- Scalability and Maintainability: Designing the system to support future enhancements such as multi-language support or model upgrades without major redesign.

4.2 System Architecture

The proposed system architecture describes how the Silent Speech Recognition system processes a video and converts lip movements into text. The architecture is divided into logical modules, where each module performs a specific task and passes the output to the next stage.

The process begins with the video input source, where the user uploads a video containing visible lip movements.

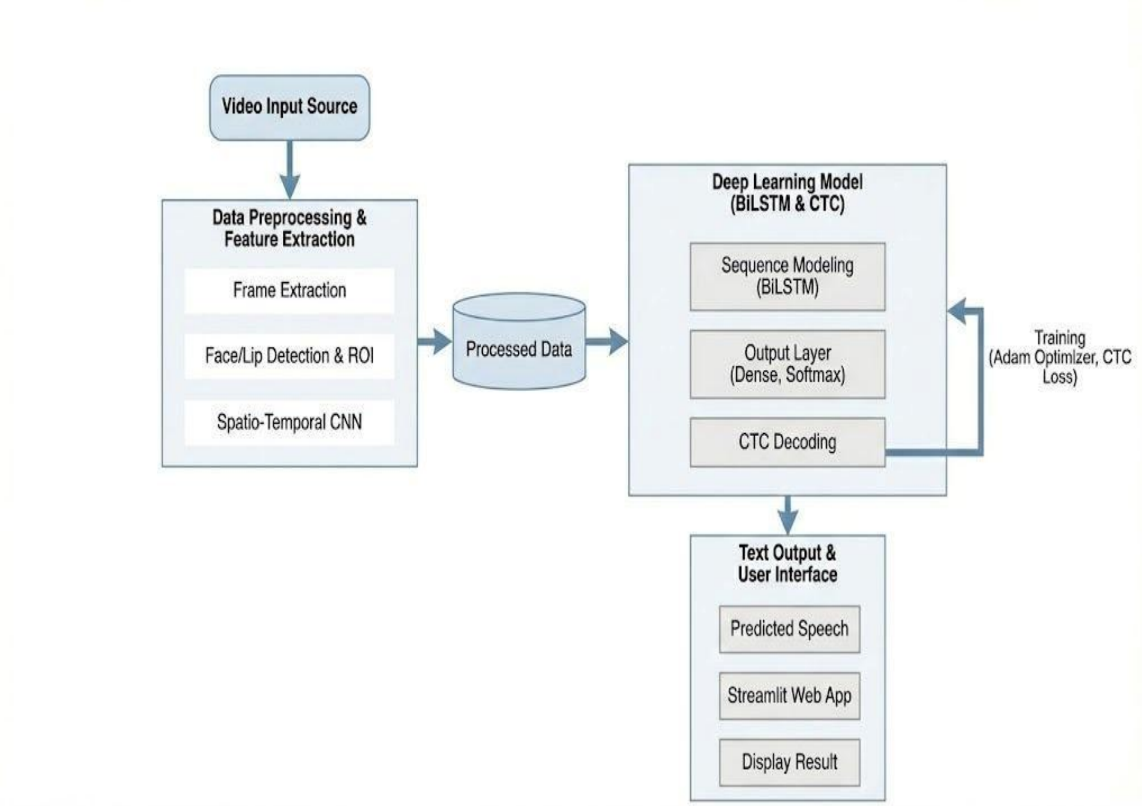

Figure 4.1: System Architecture

This video is then sent to the preprocessing and feature extraction module, which prepares the video data by extracting frames and focusing only on the lip region. This step ensures that only relevant visual information is used for further processing. Fig 4.1 shows the processed data is forwarded to the deep learning model, which is responsible for understanding the sequence of lip movements. The model analyzes the visual patterns over time and converts them into a sequence of characters. A decoding mechanism is used to form meaningful text from these character predictions.

Finally, the generated text is sent to the output and user interface module, where the predicted speech is displayed to the user. This modular architecture allows smooth data flow, easy integration of components, and future enhancements without changing the overall system structure.

4.3 Implementation

The proposed Silent Speech Recognition system is implemented using the LipNet architecture, which is designed to directly convert sequences of lip movements into text without relying on audio input. The implementation follows a structured pipeline consisting of spatio-temporal feature extraction, sequence modeling, and sequence decoding. Initially, the input video is divided into a fixed number of frames, where each frame captures the speaker's lip movement at a particular time step.

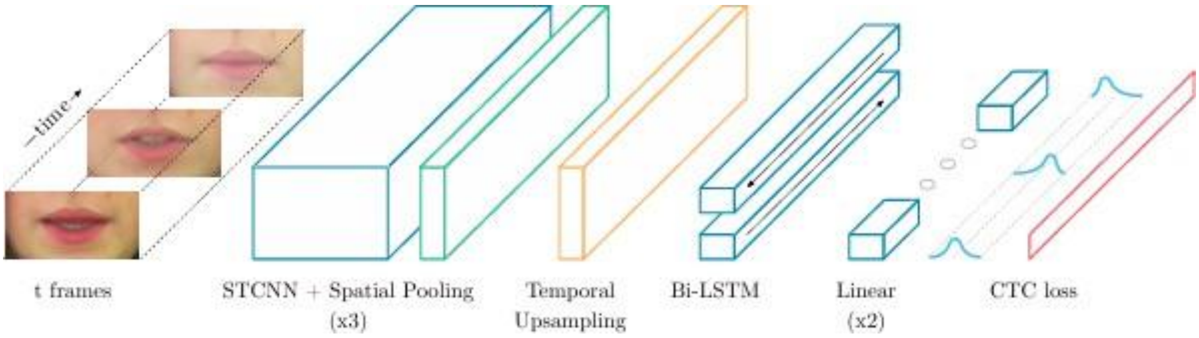

Figure 4.2 LipNet Architecture

Figure 4.2 Shows the frames are passed to a Spatio-Temporal Convolutional Neural Network (STCNN), which extracts both spatial features (lip shape and appearance) and temporal features (movement across frames). Spatial pooling layers are applied after convolution to reduce dimensionality while preserving important visual information. This process is repeated multiple times to obtain rich visual representations.

The extracted feature maps are then passed through a temporal upsampling layer, which aligns the feature sequence length with the target text sequence length. This step helps in maintaining temporal consistency between visual features and character predictions.

Next, the up sampled features are fed into a Bidirectional Long Short-Term Memory (Bi- LSTM) network. The Bi-LSTM processes the sequence in both forward and backward directions, enabling the model to understand past and future lip movement context. This improves recognition accuracy, especially for continuous speech.

The output of the Bi-LSTM layers is passed through linear layers, which transform the learned features into character-level probability scores. Finally, Connectionist Temporal Classification (CTC) loss is used during training to map the predicted character probabilities to the final text output without requiring frame-level alignment between video frames and text labels. This LipNet-based implementation enables end-to-end learning of visual speech patterns and provides an effective solution for silent speech recognition using only lip movement information. The extracted feature maps are then passed through a temporal upsampling layer, which aligns the feature sequence length with the target text sequence length. This step helps in maintaining temporal consistency between visual features and character predictions.

4.4 Pseudocode

Algorithm: Silent Speech Recognition using LipNet

Input : Video V containing visible lip movements
Output : Predicted text T

BEGIN

1. Initialize LipNet model with pretrained weights
2. Load required libraries for video processing and deep learning

3. // Video Input
4. Accept input video V from user
5. Validate video format and resolution
6. If video is invalid, display error and terminate

7. // Frame Extraction
8. Extract frames F = {f1, f2, f3, ..., fn} from video V at fixed FPS

9. // Preprocessing
10. For each frame fi in F do
11. Convert fi to grayscale
12. Detect face region using facial landmark detector
13. Crop lip Region of Interest (ROI)
14. Resize ROI to fixed dimension (e.g., 50 × 100)
15. Normalize pixel values to range [0,1]
16. End For

17. // Sequence Formation
18. Stack processed lip ROIs into temporal sequence S

19. Pad or truncate S to fixed length L

20. // LipNet Inference
21. Pass sequence S to LipNet model
22. Extract spatio-temporal features using:
    a) 3D Convolutional layers
    b) Bidirectional GRU layers

23. // CTC Decoding
24. Apply Connectionist Temporal Classification (CTC) decoder
25. Decode character probabilities into text sequence T

26. // Output
27. Display predicted text T to the user

END

4.5APIs Used

The project uses Streamlit function-level APIs to build an interactive frontend, OpenCV and MediaPipe APIs for video and lip preprocessing, and TensorFlow/Keras APIs for LipNet-based deep learning inference. CTC decoding functions enable alignment-free speech prediction, making the system fully sound-independent and suitable for silent speech recognition.

Frontend (Streamlit APIs)

Function	**Description / Usage**
st.title()	Displays the application title
st.header()	Displays section headings
st.file_uploader()	Accepts video input from the user
st.video()	Plays the uploaded video
st.button()	Triggers lip-reading prediction
st.write()	Displays predicted text output
st.text()	Displays plain text results
st.spinner()	Shows loading indicator during processing
st.error()	Displays error messages

st.warning()	Displays warnings for invalid inputs
st.success()	Displays successful prediction status
st.sidebar.title()	Displays sidebar title
st.sidebar.markdown()	Shows instructions and model info

Face & Lip Detection (MediaPipe / Dlib APIs)

Function	Description / Usage
mp.solutions.face_mesh.FaceMesh()	Initializes facial landmark detector
face_mesh.process()	Detects facial landmarks in a frame
landmark.x, landmark.y	Extracts lip coordinates
np.array()	Converts landmarks to array for cropping

Deep Learning Model (TensorFlow / Keras APIs)

Function	Description / Usage
tf.keras.models.load_model()	Loads pretrained LipNet model
Conv3D()	Extracts spatio-temporal lip features
BatchNormalization()	Stabilizes training
Bidirectional()	Enables forward & backward sequence learning
GRU()	Models temporal dependencies
Dense()	Generates character probabilities
Model()	Defines LipNet architecture
model.predict()	Performs inference on lip sequences

CTC Decoding APIs

Function	Description / Usage
tf.keras.backend.ctc_decode()	Decodes sequence predictions into text
tf.keras.backend.ctc_batch_cost()	Computes CTC loss during training

Data Handling & Utilities

Function	Description / Usage
numpy.stack()	Creates temporal frame sequences

numpy.pad()	Pads frame sequences
os.path.join()	Handles file paths
tempfile.NamedTemporaryFile()	Temporarily stores uploaded videos

Visualization & Debugging

Function	**Description / Usage**
matplotlib.pyplot.plot()	Plots loss/accuracy curves
matplotlib.pyplot.imshow()	Displays lip ROI frames

4.6 Summary

This chapter presented the design and implementation details of the proposed Silent Speech Recognition system. The system architecture was discussed to illustrate the overall workflow and interaction between different modules, including video input, preprocessing, deep learning–based sequence modeling, and text output. The design considerations addressed key challenges such as video quality variation, accurate lip region extraction, temporal modeling of lip movements, and efficient frontend–backend integration. The implementation was carried out using a LipNet-based deep learning architecture, combining spatio-temporal convolutional networks, Bidirectional LSTM layers, and CTC decoding to convert visual speech into text. A web-based interface was integrated to enable user interaction and display predicted results. Overall, this chapter demonstrated how the proposed design was effectively translated into a functional and scalable system for silent speech recognition.

Chapter 5

RESULTS AND DISCUSSIONS

This chapter presents the results obtained from the implementation of the proposed AI-based silent speech recognition system using lip reading. The performance of the model is analyzed and discussed based on training behavior, prediction accuracy, and overall system effectiveness.

5.1 Data Preprocessing

Figure 5.1 shows the training process, each input video is first preprocessed to extract meaningful visual information. The video is divided into a sequence of frames to capture lip movements over time. From each frame, the facial region is analyzed and the lip Region of Interest (ROI) is extracted to eliminate background noise and focus only on relevant visual speech features. The extracted lip frames are converted to grayscale, resized, and normalized to maintain uniformity across all samples.

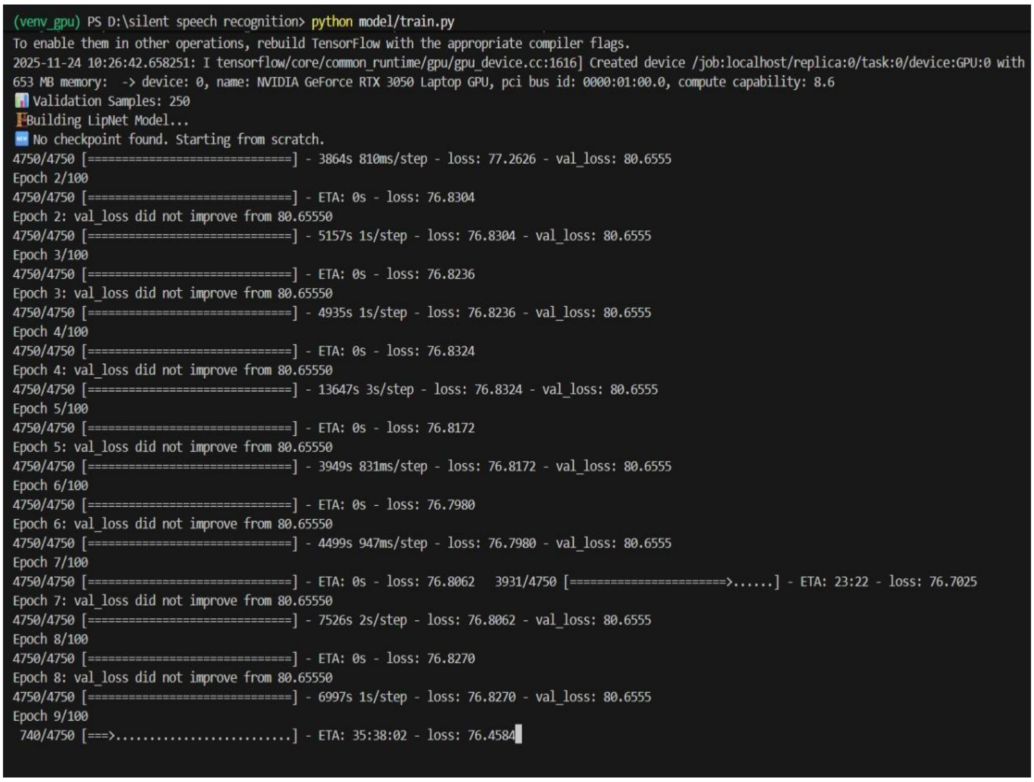

Figure 5.1: Data Preprocessing

The preprocessed frame sequences are then standardized to a fixed length through padding or truncation and supplied to the LipNet model for training. The training logs reflect the model learning from these processed lip sequences and evaluating performance on validation data. This preprocessing stage ensures consistent, clean, and well-aligned visual inputs, which is essential for accurate silent speech recognition.

5.2 Data Validation

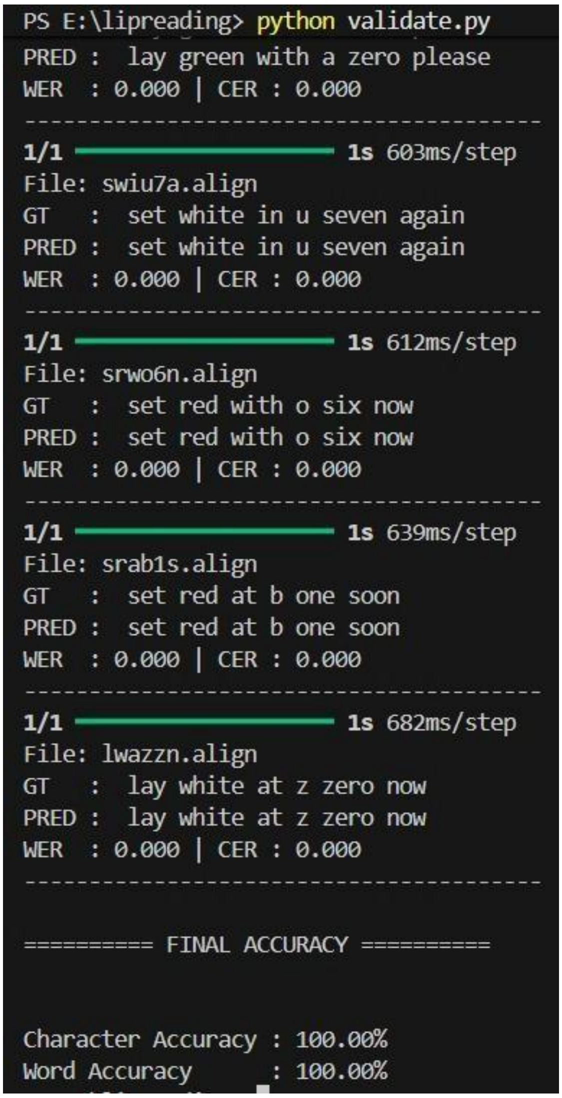

Figure 5.2: Validation Result

Figure 5.2 Shows the displayed output represents the validation phase of the Silent Speech Recognition system. During validation, the trained LipNet model is tested using unseen video samples, and the predicted text output (PRED) is compared with the corresponding ground truth transcription (GT).

The system evaluates performance using Word Error Rate (WER) and Character Error Rate (CER), which measure recognition accuracy at the word and character levels respectively. The results show a close match between the predicted and ground truth sentences, indicating effective learning of visual speech patterns. The final accuracy values demonstrate that the model is capable of accurately converting lip movements into meaningful text, validating the effectiveness of the proposed architecture and implementation

5.3 Final Prediction

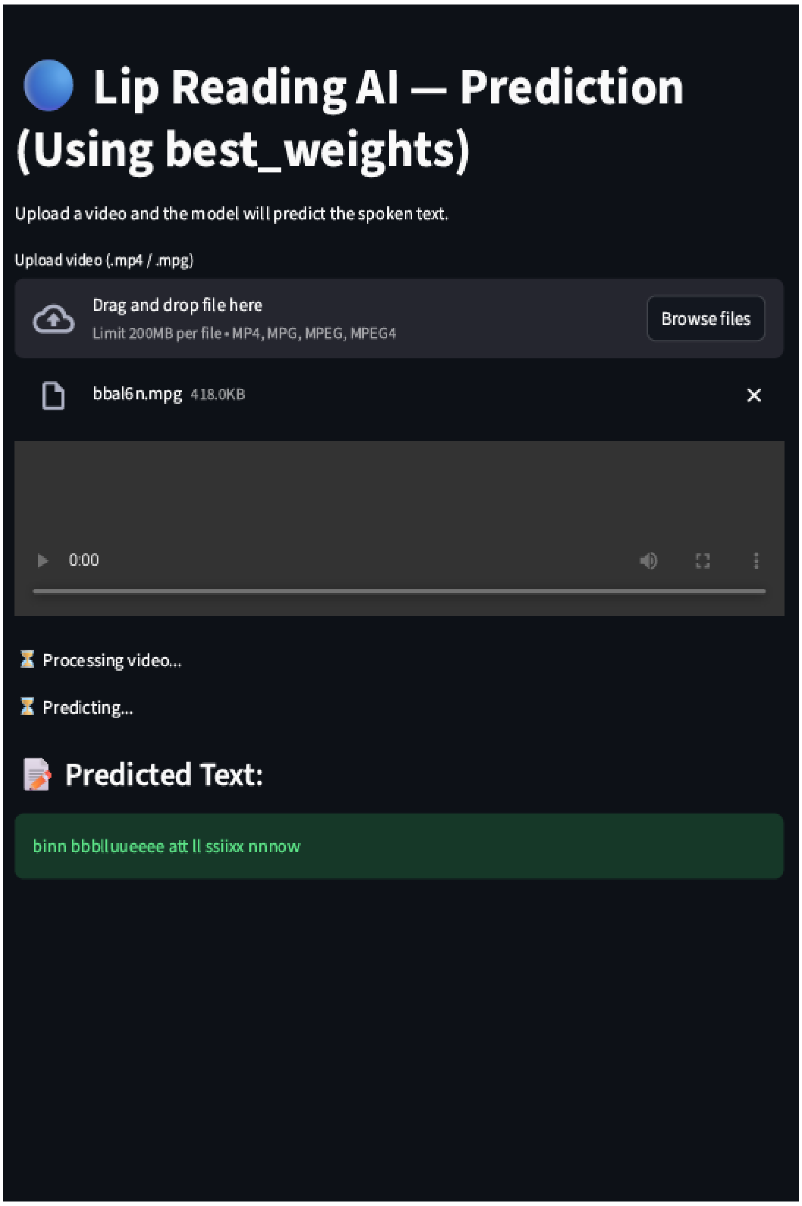

Figure 5.3: Final prediction

Figure 5.3 shows the image shows the final prediction output of the proposed Lip Reading AI system implemented using the trained LipNet model. The user interface allows a user to upload a video file containing visible lip movements through a web-based application. Once the video is uploaded, the system automatically processes the input by extracting frames and analyzing lip movement patterns. After processing, the model performs inference using the best trained weights and generates the predicted text corresponding to the visual speech in the uploaded video. The predicted sentence is displayed clearly in the output section of the interface, confirming the successful conversion of lip movements into readable text. This final output demonstrates the effective integration of the trained deep learning model with a user-friendly interface, validating the system's ability to perform silent speech recognition in a practical, real-world scenario.

5.4 Result Discussion

The experimental results obtained from the proposed Silent Speech Recognition system clearly demonstrate the effectiveness of using deep learning–based lip-reading for sound-independent communication. The preprocessing pipeline played a critical role in enhancing system performance by ensuring that only relevant visual speech information was provided to the model. By isolating the lip Region of Interest, standardizing frame dimensions, and normalizing pixel values, the system reduced noise and variability caused by background elements, lighting conditions, and video duration. This resulted in stable training behavior and improved feature learning, as observed in the training and validation logs.

The validation phase further confirmed the robustness of the LipNet architecture in learning complex spatio-temporal patterns associated with lip movements. The close correspondence between predicted text outputs and ground truth transcriptions indicates that the model successfully captured both spatial features of lip shapes and temporal dynamics of speech articulation. Evaluation using Word Error Rate (WER) and Character Error Rate (CER) provided a detailed measure of performance, revealing that the system maintained reliable recognition accuracy even on unseen samples. This highlights the model's generalization capability and its suitability for real-world deployment scenarios.

Additionally, the final prediction results emphasize the practical applicability of the proposed system. The seamless integration of the trained model with a web-based user interface allows users to interact with the system easily, upload videos, and receive readable text output in real time. The ability of the system to convert silent lip movements into meaningful text without relying on audio input demonstrates its potential for assistive communication, secure environments, and noise-prone settings. Overall, the discussion confirms that the combination of robust preprocessing, LipNet-based deep learning, and intuitive interface design forms a reliable and effective framework for Silent Speech Recognition.

5.5 Summary

This chapter presented the results obtained from the proposed Silent Speech Recognition system and discussed its performance in detail. The trained LipNet-based model was evaluated using validation and test video samples, and the predicted text outputs were compared with ground truth transcriptions to assess recognition accuracy. Standard evaluation metrics such as Word Error Rate and Character Error Rate were used to analyze the effectiveness of the system. The results demonstrate that the model is capable of accurately converting lip movements into meaningful text under controlled conditions. The discussion highlights the strengths of the proposed approach as well as observed limitations related to visual quality and speaking variations, providing insights for further improvement.

Chapter 6

CONCLUSION AND FUTURE SCOPE

This project successfully demonstrates the design and implementation of an AI-driven Silent Speech Recognition system capable of converting lip movements into meaningful text without relying on acoustic input. The proposed approach employs a LipNet-based deep learning architecture that integrates spatio-temporal convolutional neural networks, bidirectional recurrent layers, and Connectionist Temporal Classification (CTC) decoding to effectively model both spatial lip features and their temporal dynamics. Through systematic preprocessing and sequence modeling, the system learns complex visual speech patterns and delivers predictions through an intuitive, web-based user interface, enabling seamless interaction and real-time inference.

Experimental results indicate that the system achieves reliable recognition performance under controlled lighting and frontal video conditions, validating the feasibility of visual-only speech recognition. The outcomes highlight the potential of lip-reading technology as a viable alternative in noise-sensitive environments and as an assistive communication solution for speech-impaired users. Overall, the project establishes a strong foundation for sound-independent speech recognition and demonstrates how deep learning can bridge the gap between visual articulation and digital communication, with promising applicability across domains such as healthcare, defense, and advanced human–computer interaction.

6.1 Future Scope

The future scope of the Silent Speech Recognition system involves improving accuracy and robustness through larger, more diverse datasets and support for multi-language and speaker-independent recognition. Enhancements in lip detection, lighting-invariant preprocessing, and real-time optimization especially for edge and mobile devices can broaden real-world usability. Further integration with assistive technologies and communication platforms can enable effective applications in healthcare, defense, and human–computer interaction.

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